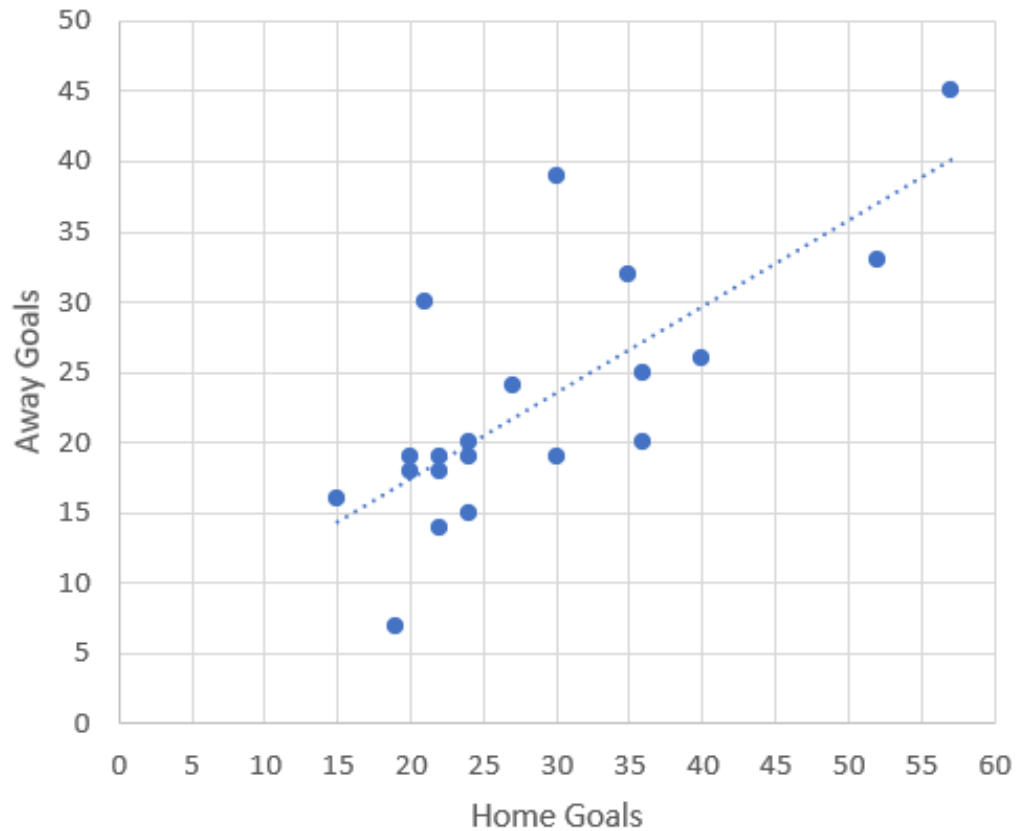


1. The diagram below shows the same data as question 2 but this time plotted as a scatter graph.



- Describe the correlation shown by this diagram and explain why this answer is not surprising.
 - Wolverhampton Wanderers scored 27 goals at home during the 2019-20 season. How many goals did they score away from home?
 - Imagine that a team's data are missing but that you know they scored 40 away goals respectively. How many home goals might you have expected them to score?
 - Why might you not use this diagram to predict the home goals scored by a team which scored 60 goals away from home?
 - At the time of writing, Chelsea are on course to score 44 goals at home this season. Why might you not use the graph above to predict how many goals they will score away?
2. The diagrams on the next page each show the proportion of people who travel to work by driving a car or van plotted against the proportion who travel to work using each of the other 10 methods. This data is taken from Method of Travel to Work for the London Boroughs in 2011.
- For each of the diagrams identify the type of correlation shown (if any).
 - What is the wrong with the following statement? "Greater levels of car use cause fewer people to use the underground, metro, light rail or tram."
 - Explain the negative correlation between car use and bicycle use
 - By referring to the diagrams in Question 11, explain why almost all of the relationships exhibit either no correlation or negative correlation.
 - Explain the reasons for the positively correlated relationship.
 - Identify any outliers.
 - Assume that there is another Local Authority which has a very high incidence, say 0.4, of people working from home. Why would you not use the below to estimate the incidence of car use?

